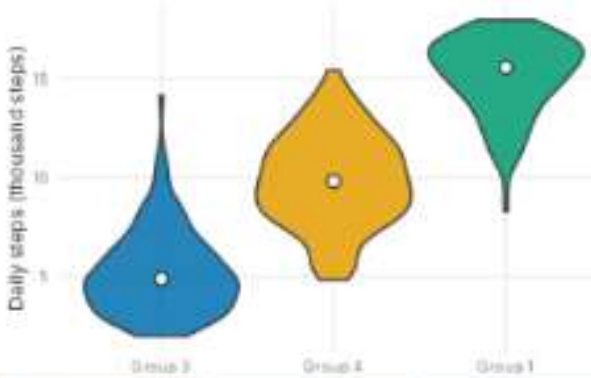
DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot C

R.No: 32 **Reg. No: 192424278** **Name: SHEIK MOHAMMED AFNAN A**

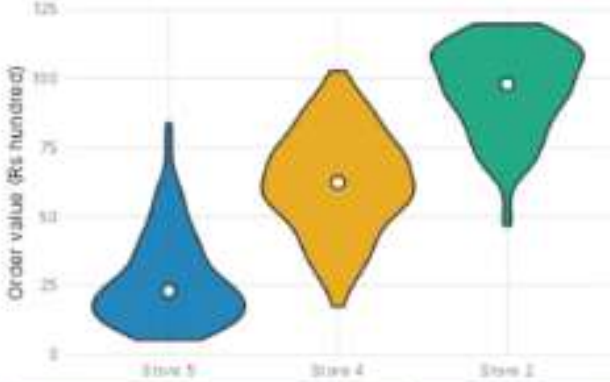
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Daily steps (thousand steps)



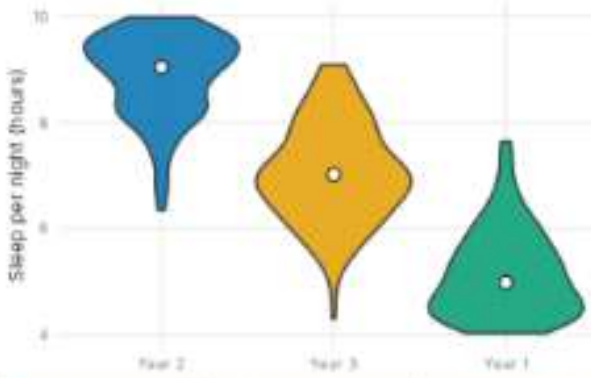
Violin	Skew (circle one)	Mean vs Median (circle)
Group 3	Right / Left / Symm.	Mean > / < / ~ Median
Group 4	Right / Left / Symm.	Mean > / < / ~ Median
Group 1	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Order value (Rs hundred)



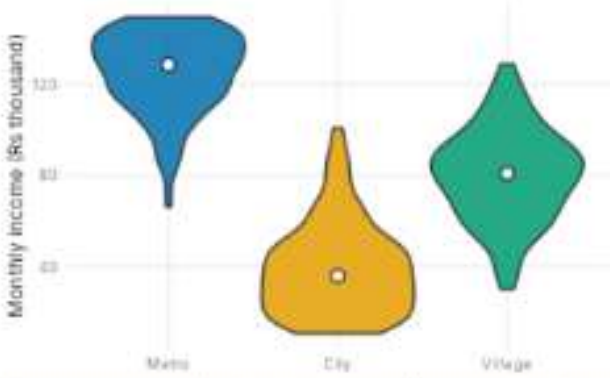
Violin	Skew (circle one)	Mean vs Median (circle)
Store 5	Right / Left / Symm.	Mean > / < / ~ Median
Store 4	Right / Left / Symm.	Mean > / < / ~ Median
Store 2	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Sleep per night (hours)



Violin	Skew (circle one)	Mean vs Median (circle)
Year 2	Right / Left / Symm.	Mean > / < / ~ Median
Year 3	Right / Left / Symm.	Mean > / < / ~ Median
Year 1	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Monthly income (Rs thousand)



Violin	Skew (circle one)	Mean vs Median (circle)
Metro	Right / Left / Symm.	Mean > / < / ~ Median
City	Right / Left / Symm.	Mean > / < / ~ Median
Village	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 4, violin "Metro" is not symmetric. Towards which side is its tail, and which is the better measure of a typical monthly income here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

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192424278 (32)

ACTIVITY - 4

08/10/2026
DSA0601 - DHEV

Activity : Identify skewness using violin plots

Part A -

Rule used :-

- Right - Skewed : Longer tail toward higher values → mean > Median
- Left - Skewed : Longer tail toward lower values → Mean < Median
- Symmetric : both sides are approximately equal → Mean ≈ Median

The White Circle represents the Median.

Chart 1: Daily Steps (thousand steps)

violin	Skew	Mean vs Median	Reason
Group 3	Right skewed	Mean > Median	Longer tail extends toward higher step values
Group 2	symmetric	Mean ≈ Median	Distribution is approximately balanced on both sides
Group 1	Left skewed	Mean < Median	Longer tail extends toward lower step values

Chart 2: order value (Rs hundred)

violin	Skew	Mean vs Median	Reason
Store 5	Right Skewed	Mean > Median	Longer upper tail towards larger order values
Store 4	Symmetric	Mean ≈ Median	The distribution is nearly balanced around the center
Store 2	Left-Skewed	Mean < Median	Long lower tail toward smaller order values

Chart - 3

violin	Skew	Mean vs Median	Reason
Year 2	Left Skewed	Mean < Median	Longer tail extends towards lower sleep hours
Year 3	Left Skewed	Mean < Median	Lower side has a slightly longer tail than the upper side
Year 1	Right Skewed	Mean > Median	Longer tail extends towards higher sleep hours

Chart 4 : Monthly Income (Rs Thousand)

Violin	Skew	Mean vs Median	Reason
Metro	Left skewed	Mean < Median	Long tail extends towards lower income values
City	Right skewed	Mean > median	long upper tail extends toward higher income values
Village	Symmetric	Mean ≈ Median	Distribution is approximately balanced around the center

Part B

B1) Ans: The Metro violin is left-skewed, because its tail extends toward the lower income values

The Median is better measure of typical monthly income

Reason: The long lower tail can contain usually low income values that pull the mean downward, while the median is less affected by extreme values and skewness

~~Final Answer~~

B2).

Ans: ggplot (df, aes(x = group, y = value)) + geom_violin() + stat_summary(fun = median, geom = "point")

Explanation:-

ggplot (df, aes(...)) → uses the df data frame
group → appear on the x-axis
value → appears on y-axis
geom_violin → creates violin plots
stat_summary (fun = median, geom = "point") → displays the Median as a point